

HDOC Day-9

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Batch: B-16

Program: B.Tech CSE

Question 1

Aim: Swap two numbers using and without a third variable

Algorithm

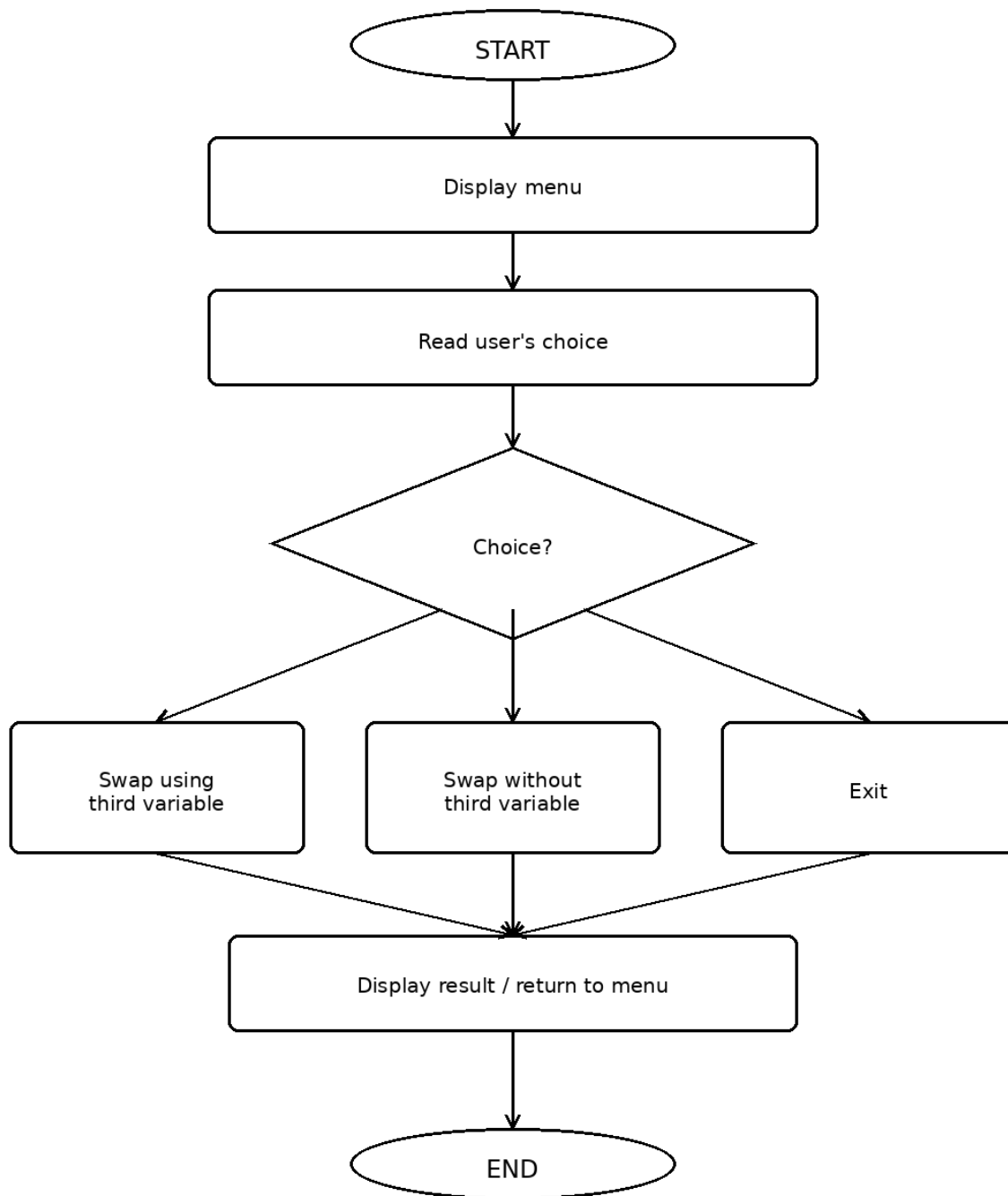
1. Start the program.
2. Display a menu with options: swap using a third variable, swap without a third variable, and exit.
3. Read the user's choice.
4. If choice is 1, input two numbers, display their values before swapping, swap them using a third variable, and display the values after swapping.
5. If choice is 2, input two numbers, display their values before swapping, swap them without using a third variable, and display the values after swapping.
6. If choice is 3, terminate the program.
7. Otherwise, display an invalid-choice message.
8. Repeat the menu until the user chooses Exit.
9. Stop.

Pseudocode

```
BEGIN
REPEAT
    DISPLAY swap menu
    READ choice
    IF choice = 1 THEN
        INPUT a, b
        DISPLAY values before swapping
        temp ← a
        a ← b
        b ← temp
        DISPLAY values after swapping
    ELSE IF choice = 2 THEN
        INPUT a, b
        DISPLAY values before swapping
        a ← a + b
        b ← a - b
        a ← a - b
        DISPLAY values after swapping
    ELSE IF choice is not 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

```
UNTIL choice = 3  
END
```

Flowchart



C Program

```
#include <stdio.h>

int main() {
    int choice;
    float a, b, temp;

    do {
        printf("\n--- SWAP TWO NUMBERS ---\n");
        printf("1. Swap using third variable\n");
        printf("2. Swap without third variable\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                printf("Before swapping: a = %.2f, b = %.2f\n", a, b);
                temp = a;
                a = b;
                b = temp;
                printf("After swapping: a = %.2f, b = %.2f\n", a, b);
                break;

            case 2:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                printf("Before swapping: a = %.2f, b = %.2f\n", a, b);
                a = a + b;
                b = a - b;
                a = a - b;
                printf("After swapping: a = %.2f, b = %.2f\n", a, b);
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	Choice 1; a=10, b=20	a=20, b=10	a=20, b=10	PASS
TC2	Choice 2; a=15, b=30	a=30, b=15	a=30, b=15	PASS
TC3	Choice 9	Invalid choice message	Invalid choice message	PASS

Program Output Screenshots

Screenshot 1

```
(kali㉿kali)-[~/Desktop]
$ gcc hdoc1.c -o hdoc1 -lm

(kali㉿kali)-[~/Desktop]
$
```

Screenshot 2

```
(kali㉿kali)-[~/Desktop]
$ ./hdoc1

===== MENU =====
1. Swap using third variable
2. Swap without third variable
3. Exit
Enter your choice: 1

Enter two numbers: 10
20

Before swapping:
a = 10
b = 20

After swapping:
a = 20
b = 10
```

Screenshot 3

```
===== MENU =====  
1. Swap using third variable  
2. Swap without third variable  
3. Exit  
Enter your choice: 2  
  
Enter two numbers: 15  
30  
  
Before swapping:  
a = 15  
b = 30  
  
After swapping:  
a = 30  
b = 15
```

Screenshot 4

```
===== MENU =====  
1. Swap using third variable  
2. Swap without third variable  
3. Exit  
Enter your choice: 3  
  
Exiting Program ...
```

Question 2

Aim: Find the sum of the first n natural numbers and the sum of squares of the first n natural numbers

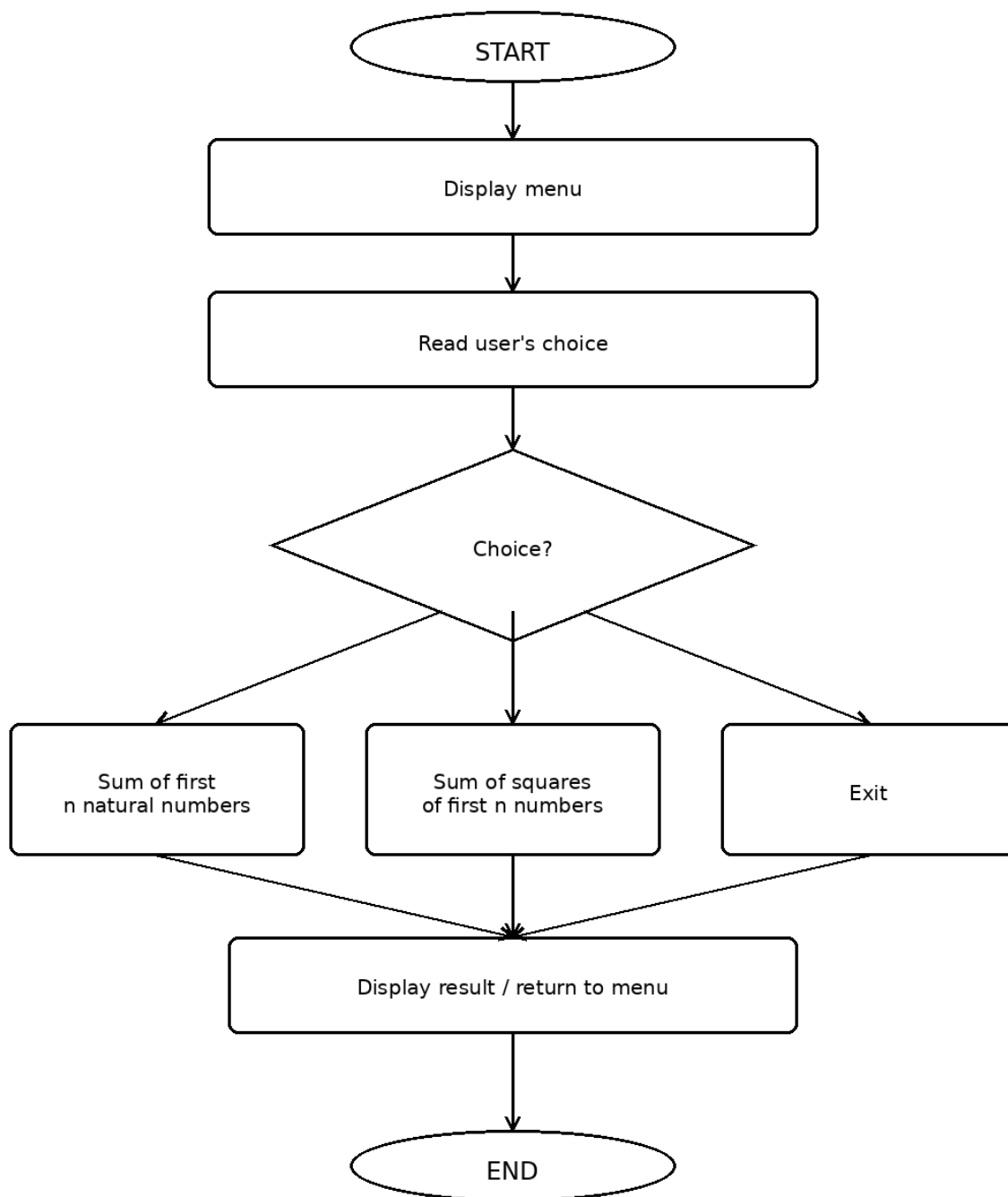
Algorithm

1. Start the program.
2. Display a menu with options: sum of first n natural numbers, sum of squares of first n natural numbers, and exit.
3. Read the user's choice.
4. If choice is 1, input n , verify that n is positive, and calculate $\text{Sum} = n(n + 1)/2$.
5. If choice is 2, input n , verify that n is positive, and calculate $\text{Sum of Squares} = n(n + 1)(2n + 1)/6$.
6. Display the calculated result.
7. If choice is 3, terminate the program.
8. Otherwise, display an invalid-choice message.
9. Repeat the menu until the user chooses Exit.
10. Stop.

Pseudocode

```
BEGIN
REPEAT
    DISPLAY calculation menu
    READ choice
    IF choice = 1 THEN
        INPUT n
        IF n > 0 THEN
             $\text{sum} \leftarrow n \times (n + 1) / 2$ 
            DISPLAY sum
        END IF
    ELSE IF choice = 2 THEN
        INPUT n
        IF n > 0 THEN
             $\text{squareSum} \leftarrow n \times (n + 1) \times (2n + 1) / 6$ 
            DISPLAY squareSum
        END IF
    ELSE IF choice is not 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

Flowchart



C Program

```
#include <stdio.h>
```

```

int main() {
    int choice, n, i;
    long long sum = 0, squareSum = 0;

    do {
        printf("\n--- SUM CALCULATIONS ---\n");
        printf("1. Sum of first n natural numbers\n");
        printf("2. Sum of squares of first n natural numbers\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter n: ");
                scanf("%d", &n);
                if (n > 0) {
                    sum = (long long)n * (n + 1) / 2;
                    printf("Sum of first %d natural numbers = %lld\n", n, sum);
                } else {
                    printf("Please enter a positive natural number.\n");
                }
                break;

            case 2:
                printf("Enter n: ");
                scanf("%d", &n);
                if (n > 0) {
                    squareSum = (long long)n * (n + 1) * (2 * n + 1) / 6;
                    printf("Sum of squares of first %d natural numbers = %lld\n",
                        n, squareSum);
                } else {
                    printf("Please enter a positive natural number.\n");
                }
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}

```

Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	Choice 1; n=5	15	15	PASS
TC2	Choice 2; n=5	55	55	PASS
TC3	Choice 1; n=-2	Positive-number message	Positive-number message	PASS

Program Output Screenshots

Screenshot 1

```
(kali㉿kali)-[~/Desktop]
$ gcc hdoc1.c -o hdoc1 -lm

(kali㉿kali)-[~/Desktop]
$
```

Screenshot 2

```
===== MENU =====
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1

Enter the value of n: 5

Sum of first 5 natural numbers = 15
```

Screenshot 3

```
===== MENU =====
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2

Enter the value of n: 5

Sum of squares of first 5 natural numbers = 55
```

Screenshot 4

===== MENU =====

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit

Enter your choice: 3

Exiting Program...

Question 3

Aim: Calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time

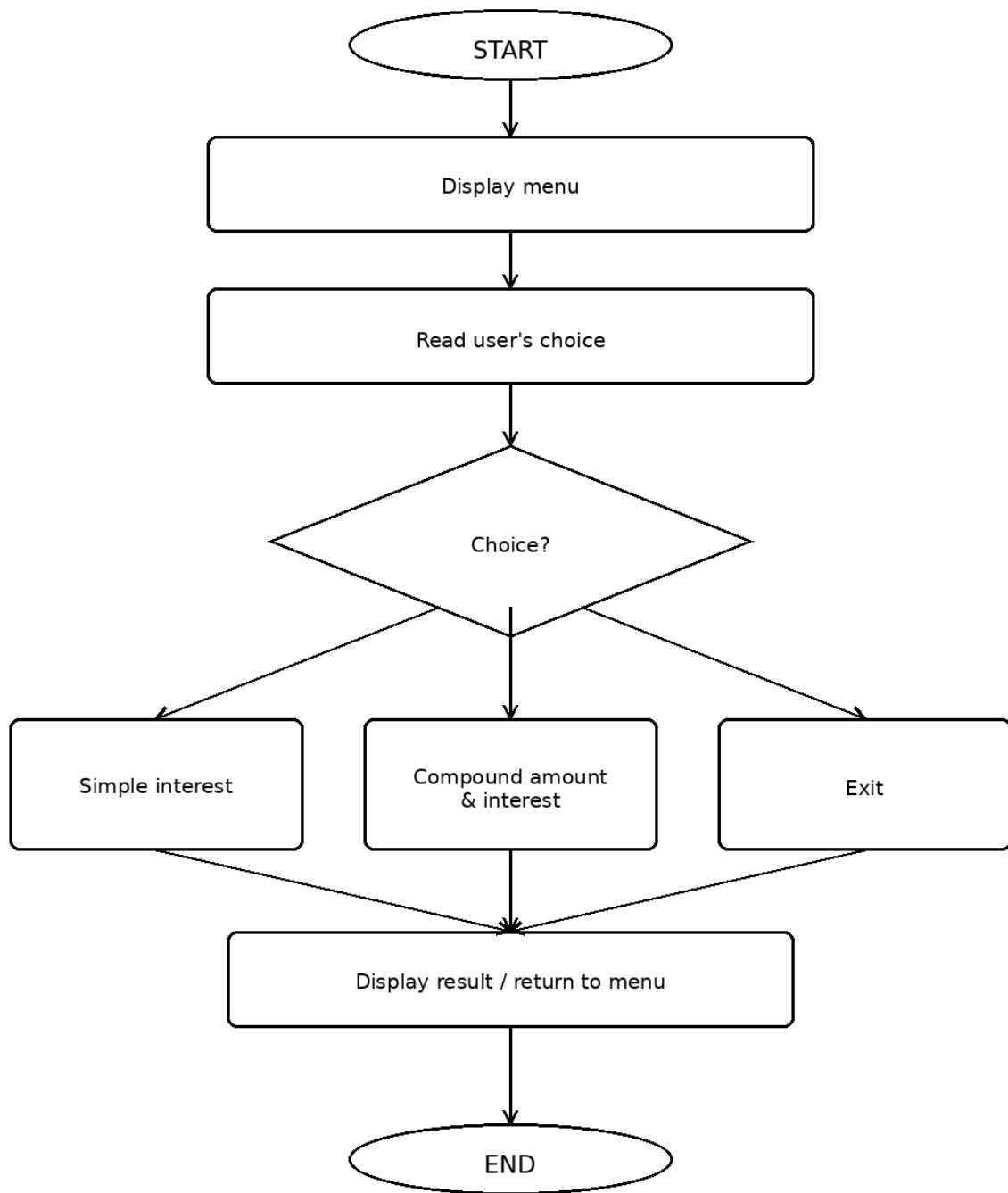
Algorithm

1. Start the program.
2. Display a menu with options: simple interest, compound amount and compound interest, and exit.
3. Read the user's choice.
4. If choice is 1, input principal, rate, and time; calculate $SI = (P \times R \times T)/100$ and total payable amount = $P + SI$.
5. If choice is 2, input principal, rate, and time; calculate compound amount $A = P(1 + R/100)^T$ and compound interest $CI = A - P$.
6. Display the required interest and payable amount.
7. If choice is 3, terminate the program.
8. Otherwise, display an invalid-choice message.
9. Repeat the menu until the user chooses Exit.
10. Stop.

Pseudocode

```
BEGIN
REPEAT
    DISPLAY interest menu
    READ choice
    IF choice = 1 THEN
        INPUT P, R, T
         $SI \leftarrow (P \times R \times T) / 100$ 
         $Total \leftarrow P + SI$ 
        DISPLAY SI and Total
    ELSE IF choice = 2 THEN
        INPUT P, R, T
         $A \leftarrow P \times (1 + R/100)^T$ 
         $CI \leftarrow A - P$ 
        DISPLAY A, CI and Total Payable Amount
    ELSE IF choice is not 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

Flowchart



C Program

```
#include <stdio.h>
```

```

#include <math.h>

int main() {
    int choice;
    double principal, rate, time, simpleInterest;
    double compoundAmount, compoundInterest;

    do {
        printf("\n--- INTEREST CALCULATIONS ---\n");
        printf("1. Calculate Simple Interest\n");
        printf("2. Calculate Compound Amount and Compound Interest\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter principal, rate and time: ");
                scanf("%lf %lf %lf", &principal, &rate, &time);

                simpleInterest = (principal * rate * time) / 100;
                printf("Simple Interest = %.2lf\n", simpleInterest);
                printf("Total Payable Amount = %.2lf\n",
                    principal + simpleInterest);
                break;

            case 2:
                printf("Enter principal, rate and time: ");
                scanf("%lf %lf %lf", &principal, &rate, &time);

                compoundAmount = principal *
                    pow((1 + rate / 100), time);
                compoundInterest = compoundAmount - principal;

                printf("Compound Amount = %.2lf\n", compoundAmount);
                printf("Compound Interest = %.2lf\n", compoundInterest);
                printf("Total Payable Amount = %.2lf\n", compoundAmount);
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}

```

Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	Choice 1; P=10000, R=10, T=2	SI=2000; Total=12000	SI=2000; Total=12000	PASS
TC2	Choice 2; P=10000, R=10, T=2	A=12100; CI=2100	A=12100; CI=2100	PASS
TC3	Choice 9	Invalid choice message	Invalid choice message	PASS

Program Output Screenshots

Screenshot 1

```
(kali㉿kali)-[~/Desktop]
$ gcc hdoc3.c -o hdoc3 -lm

(kali㉿kali)-[~/Desktop]
$
```

Screenshot 2

```
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 1

Enter Principal: 1000
Enter Rate (%): 10
Enter Time (years): 2

Simple Interest = 200.00
```

Screenshot 3

```
===== MENU =====  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
5. Exit  
Enter your choice: 2  
  
Enter Principal: 1000  
Enter Rate (%): 10  
Enter Time (years): 2  
  
Compound Amount = 1210.00
```

Screenshot 4

```
===== MENU =====  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
5. Exit  
Enter your choice: 3  
  
Enter Principal: 1000  
Enter Rate (%): 10  
Enter Time (years): 2  
  
Compound Interest = 210.00
```

Screenshot 5

```
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 4

Enter Principal: 1000
Enter Rate (%): 10
Enter Time (years): 2

Total Payable Amount = 1210.00
```

Screenshot 6

```
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 5

Exiting Program...

(kali@kali)-[~/Desktop]
$
```